

42088 – Industrial Project

Use Cases

Milestone 2 – Elaboration

Project name:	Automatic, universal push-button actuator, equipped with force gauge
Customer:	Altice Labs
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Introduction

This paper describes the software requirements and use cases for the automated button actuator system, which is intended to evaluate the functionality of various buttons on devices under test (DUTs). The system's objective is to automate button testing by delivering precise force to each button and assuring consistent, repeatable interactions. The system will be operated by an API and a web interface, allowing for remote control and data collecting. It will also generate useful testing data such as force exerted and device reaction.

The Software Requirements Specification (SRS) specifies the software components of the automated button actuator system. It describes how the system will communicate with DUTs using remote commands, capture force data, validate button answers, and provide a user-friendly interface for test engineers. This document will also describe the use scenarios for the DUT's three buttons (Reset, WPS, and Info) and the intended results for each.

Purpose

This article serves as an in-depth guide for Altice Labs test engineers on how to operate the Automated Button Actuator with Force Measurement system. The setup, operation, and button testing of the actuator system are covered in this article, along with steps for starting tests, tracking force data, and reading test outcomes. It is meant to guarantee that end users can efficiently use the features of the system to perform reliable, accurate, and efficient button testing across a range of devices.

Scope

The Automated Button Actuator with Force Measurement system's functionality is fully described in this document. It contains thorough use cases and functional requirements that together provide a thorough manual for use the system for testing applications. It includes everything Altice Labs test engineers need to know to set up, start, and keep an eye on button testing using the actuator system. Instructions for remote control, data collecting, and result interpretation are included to guarantee precise, reliable testing on a variety of devices.

System Overview

The purpose of this project is to automate button testing, which includes pressing buttons, checking their operation, and determining the force required to activate them. Traditional manual testing can be inconsistent and time-consuming, making remote and exact testing difficult, particularly for devices with numerous buttons. Automating this procedure improves precision, consistency, and dependability, allowing Altice Labs to test buttons remotely and more accurately across a variety of DUTs. This technology increases testing speed, data accuracy, and workflow efficiency, benefiting applications other than Altice Labs by cutting labour costs and enabling data-driven advances in product design, reliability, and manufacturing. The system will be adjustable to various button kinds and sizes, making it a versatile, scalable solution that matches the demands of modern testing environments.

References

Previously drafted documents that can provide a good insight into the project:

- **Project Vision Document:** Contains the overall vision, purpose, customer market and high-level objectives of the Automated Button Actuator project.
- **Project Plan Document:** Outlines the project's management, timeline, resources available and risks involved.

Definitions

API (Application Programming Interface): An API is a set of rules and protocols that enable various software components to communicate and interact with one another. In this project, the API acts as a link between the user interface and the automated button actuator system, allowing remote control and monitoring of the testing process. Test engineers can use the API to start tests, change parameters (such as force settings), and receive data from the system, all from a connected device. This allows for seamless engagement with the actuator system without requiring direct, physical operation, making testing more efficient and accessible.

DUT: Device under test.

Use Cases

Name	UC-1: Test Definition
Summary	Definition of the characteristics of the test that will be performed.
Rationale	DUT's come with various types of buttons with different sizes, stiffness and functionalities that alter based on the time it is pressed. Keeping in mind that many aspects are unique to each test; in order to obtain reliable results, the user must set the parameters that best fit the test he wishes to run.
Users	Test Engineers
Preconditions	The webpage is open, in a default state and ready to be configured.
Basic Course of Events	- The user should specify: - What button will be tested; - How long the actuator should hold the button down (short press or long press); - Number of times to be pressed; - Maximum force that can be applied to the button; - Interval between actuations.
Alternative Paths	- In step 1, if the user does not input any data, the system will proceed to use default values instead. - In step 1. d, if the user enters an imposed force value that is close to or exceeds the value set as the maximum acceptable value or an out-of-range value, the system will display a warning message and suggest acceptable ranges.

Postconditions	Test is now defined, and the user can click on the start button on the Web interface to initiate it.
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Name	UC-2: Test Realization
Summary	Realization of the test.
Rationale	Manual button testing has always been an inefficient, repeating time-consuming task for Test Engineers. This Use-Case immediately follows the previous UC-1.
Users	Test Engineers
Preconditions	The webpage already contains information on the test characteristics.
Basic Course of Events	1. The test engineer holds the clamp on the DUT in an adequate position to push the button; 2. The test engineer must ensure that the DUT is powered on and ready to respond to button presses 3. On the webpage, the test engineer clicks on the start button; 4. The test engineer waits while the automaton pushes the button; 5. Once the operation is over, graphics and results regarding applied force over time are shown on the webpage for user information.
Alternative Paths	1. In step 1, to test a reset button, the user should additionally add the needle shaped accessory; 2. In step 4, if the clamp isn't positioned appropriately it may lead to the button not be pushed correctly or not pushed at all. The user should abort the test using the web interface and repeat from step 1.
Postconditions	The test has ended, and the results are available on the website.

Name	UC-3: Accessing older tests and statistics
Summary	Viewing older tests and overall statistics
Rationale	Statistics are important to extract meaningful patterns to maximize corporate profits.
Users	Company management teams
Preconditions	Webpage is open.
Basic Course of Events	1. The user accesses the Statistics tab in the Web interface; 2. The page responds by plotting the overall tests done and a search bar; 3. The user filters the parameter through which they would like to access past tests.
Alternative Paths	1. In step 2 if there's a failure in retrieving data due to a system error or loss of connection and the system displays the message "Unable to retrieve test data. Please check your connection or try again later." the user can retry the action, refresh the page; 2. In step 2 if only part of the data loads (e.g., due to server latency or heavy data volume) the system will display a loading indicator and will prompt the user with the message: "Loading data, please wait."

	If loading times out, the user will be given the option to retry or narrow the search parameters. 3. In step 3 in case the system displays the message “No results found matching the selected parameters.”, the user can modify the filter criteria or clear the filters to view all available tests;
Postconditions	A table is shown with the tests performed on that period/parameter.

Functional Requirements

Name	FR-1: Data input webpage
Summary	Platform that allows the users to configure and start the test.
Rationale	This web page is what allows the user to customize the test he wants to perform to the best fit and start it.
Requirements	• User must be able to input the test configuration data; • System should validate the inputted data and only then allow to click on the start button.
References	UC-1: Test Definition

Name	FR-2: Results webpage
Summary	Platform that allows the users to see the result of the performed test.
Rationale	This web page is what allows the user to see the outputs of the system.
Requirements	System should show: • Basic data about the gateway (model, memory capacity, CPU...); • If the pressed button is functional; • Plot of the force over the time.
References	UC-2: Test Realization

Name	FR-3: Statistics webpage
Summary	Platform that allows the users to see the result of previous performed tests.
Rationale	This web page is what allows the user to see statistics of all tests made over the time.
Requirements	System should show a list of the more recent tests for easier accessibility; System should allow the user to search for specific tests based on: • A time period; • A specific type of button; System should also show plots of the number of tests over time and the overall button functionality.
References	UC-3: Accessing older tests and statistics

Name	FR-4: Compatibility of the DUT with the clamp
Summary	Verify DUT dimensions and button alignment before testing.
Rationale	The user must ensure before starting the test that the DUT meets the dimensions allowed and for which the product was developed, otherwise, it

	won't have the stability or precision to press buttons, and the purpose of the product won't be fulfilled.
Requirements	The test operator must first inspect the device in hand, in particular its design and should pay attention to conditions like: • The width of the device must not exceed the maximum opening of the claw, so that it can be attached to the device; • The DUT's surface must be smooth and stable enough for the clamp to attach securely. • The buttons on the device must be perpendicular to the surface of the device so that the user can position the clamp intuitively and directly to start the test.
References	UC-2: Test Realization

Name	FR-5: Automaton to press the buttons
Summary	Equipment that attaches to the DUT, allowing the actuator to push the buttons
Rationale	This is the equipment that allows the realization of automated and remote testing.
Requirements	System should do: • Receive the configurations introduced in the webpage; • Perform the pressing of buttons accordingly the test definition;
References	UC-2: Test Realization

Name	FR-6: Sensor to measure the applied force
Summary	Sensor that measures the necessary force to push the buttons and transmits that data to the website.
Rationale	This is the system that allows the users to study the forces applied in the buttons.
Requirements	The system should do: • Sensor measures the force applied; • Sensor measures for how long the force is applied; • Send that information to the website.
References	UC-2: Test Realization